1. Both are voiced and bilabial sounds. “m” is nasal and “b” is plosive.

2. Learning rate in SGD modulates the speed of learning from mistakes to approach the goal of learning. Stiffness in spring determines how fast a spring returns to its equilibrium status. When learning rate is high, learning in SGD is fast but may be inaccurate; low, learning is slow but accurate. High/low stiffness leads to faster/slower return of spring to equilibrium. (Not necessary but acceptable: when the learning rate is high, we may observe oscillations in SGD learning process)

3. Perturbation theory (mass ⬆️frequency ⬇️; stiffness ⬆️frequency ⬆️) The two sides of the the basilar membrane are different in mass and stiffness. The longer side is light and floppy (low mass and high stiffness) such that it captures high frequency sounds; the short side is heavy and dull (high mass and low stiffness) such that it captures low frequency sounds. Both sides are connected to the brain by neurons, which is how the brain receives such signals of whether we heard a low f or high f sound.

4. a) is like a single-spring system. b) is like a mass-spring system

5. The exhaustive algorithm is slow although accurate, which makes it inefficient in speed of solving a problem.

6. They can be distinguished by two of the three defining features of consonants, which are voicing; place of articulation. P, T, K are voiceless. B, D, G are voiced. P, B are bilabial. T, D are alveolar. K, G are velar.

7. Closure is a way to test human knowledge in completing things. GPT models are trained to complete sentences.

8. Knowledge is represented as weights in NNs.

9. Speed-accuracy trade-off: speed ⬆️accuracy ⬇️; speed ⬇️accuracy ⬆️)

10. Fake it till you make it. SGD starts out with random weights, which may well be wrong; but through the training it eventually approximates the correct solution.

11. Probability distribution of all possible outcomes. Choose the highest probability one(s).
12. No, they add up to more than 1
13. No
14. Vowel height and frontness/backness
15. Mass-spring system; speed sounds; temperature in the room regulated by AC; economic development; SGD with a high learning rate
16. Yes. X will move towards 0 (equilibrium) and v will increase. When x reaches the positive maximum, v becomes 0, and x starts to return to equilibrium again, but v will increase again. So on and so forth.

17. X goes up towards 0 (equilibrium), first fast, then gradually slower. X changes much faster.

18. The first minus is to make sure the learning moves towards the opposite direction of the loss. If the loss is positive (suggesting the current estimate is too much) then the next estimate should decrease one. If the loss is negative (suggesting the current estimate is too little) then the next estimate should increase from the current one.

19. Between the two neurons that are connected, get the Inner product between activation of the previous neuron and the weight, and then add the inner products up for all the neurons that are connected to the same neuron. This calculation repeats itself from layer to layer, and in the end the output is a probability distribution of all possible outcomes and the sum of the probabilities add up to 1.

20. [0,0,1,0,0] since the 2th position is the third number in a Python list